

D634

Health Sciences Capstone

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A health sciences student who works part-time at a busy pharmacy writes this question: “In adults over 65 living in assisted living, does a twice-weekly balance class reduce falls over six months?” Her capstone mentor recently moved from another state. Which PICO element is missing?

- A. Outcome
- B. Intervention, because the pharmacy setting is not named
- C. Comparison
- D. Nothing is missing, because the six-month time frame completes it

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which of these is a well-formed PICO question?

- A. What is type 2 diabetes?
- B. In adults with type 2 diabetes, does telehealth coaching compared with standard clinic visits lower A1C levels?
- C. Does telehealth coaching work?
- D. Telehealth coaching always lowers A1C in every adult.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A student chose her capstone topic after her grandmother caught an infection during a hospital stay. Her school library subscription covers four databases. Which source gives the strongest evidence that a hand-hygiene program reduces infections?

- A. A single large randomized controlled trial
- B. Whichever study is newest, since recency always beats design
- C. A systematic review and meta-analysis of randomized trials
- D. A case-control study comparing infected and uninfected patients

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

For a question about whether a treatment works, which source ranks lowest on the evidence hierarchy?

- A.** Expert opinion
- B.** A systematic review of RCTs
- C.** A randomized controlled trial
- D.** A cohort study

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A student at a university with a large online program is researching teen vaping. Her roommate recommends posts by a popular wellness influencer. Which source is most appropriate for an evidence-based capstone?

- A.** A peer-reviewed journal article from the past three years reporting a study of teen vaping
- B.** A 20-year-old textbook chapter on teen tobacco use
- C.** A fact sheet from an e-cigarette manufacturer
- D.** The influencer's most-viewed post on vaping

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A one-time survey of 2,000 students at a university known for its football team finds that students who sleep less report higher stress. The survey was funded by a campus wellness grant. What conclusion does the design support?

- A.** No conclusion, because 2,000 students is too few
- B.** Higher stress causes less sleep
- C.** Less sleep causes higher stress
- D.** Less sleep and higher stress are associated, but the design cannot show cause or which came first

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A student who volunteers at a food bank with a new walk-in freezer wants to understand how rural caregivers experience arranging transport to dialysis. Her professor prefers statistical studies. Which approach fits the question?

- A. A randomized trial of transport vouchers
- B. A survey scoring satisfaction from 1 to 5
- C. A statistical study, because that is what the professor prefers
- D. Qualitative in-depth interviews with caregivers

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A student wants to know whether average patient wait times differ between two clinics. Which approach fits?

- A. Either approach, since every method fits every question
- B. A quantitative comparison of mean wait times
- C. Focus groups on how patients feel about waiting
- D. Phenomenological interviews about waiting

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A capstone student at a hospital that recently opened a new cancer wing wants to survey patients about their care. She has analysis software and a faculty mentor with 30 years' experience. Before collecting any patient data, what must she obtain?

- A. Sign-off from the hospital's marketing office
- B. IRB review or approval, plus informed consent from each participant
- C. Her mentor's approval only
- D. Nothing, because surveys never need review

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A student writing her literature review during a semester when she also started a new job lists 15 articles one by one, with a paragraph on each. Her instructor says the review needs synthesis. What should she do?

- A.** Remove any study that disagrees with her position
- B.** Add ten more articles so the review is longer
- C.** Reorder the article summaries by publication date
- D.** Organize the studies by theme, compare their findings, and identify gaps

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

Using the library's new discovery interface, a student searches "obesity AND children AND school lunch" and gets 40 results. She wants more results, including articles that use the word "pediatric." What should she change?

- A.** Search (children OR pediatric) AND obesity AND school lunch
- B.** Remove all operators, since operators always reduce results
- C.** Add "AND pediatric" to the search
- D.** Add "NOT pediatric"

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A search for "telehealth OR telemedicine" returns 900 articles. "Telehealth" alone returns 500 and "telemedicine" alone returns 600. How many articles contain both terms?

- A.** 400
- B.** 1,100
- C.** 200
- D.** 100

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A health sciences student ~~who works part time at a busy pharmacy~~ writes this question: "In adults over 65 living in assisted living, does a twice-weekly balance class reduce falls over six months?" ~~Her capstone mentor recently moved from another state.~~ Which PICO element is missing?

Target word: missing element **Decides:** No comparison group is named.

A	Confusion set	Falls are the outcome and are named.
B	Camouflage	The pharmacy job is irrelevant; the balance class is the intervention.
C	Correct	The question never states what the balance class is compared with, such as usual activity.
D	Adjacent category	Time is the optional T in PICOT, not the comparison.

2. Answer B Concept 1 • transfer

Which of these is a well-formed PICO question?

Target word: well-formed PICO **Decides:** All four parts present.

A	Adjacent category	A background question, not a focused clinical question.
B	Correct	It names population, intervention, comparison, and outcome.
C	Dropped element	Drops the population, comparison, and outcome.
D	Absolute language	A sweeping claim, not a researchable question.

3. Answer C Concept 2

A student chose her capstone topic ~~after her grandmother caught an infection during a hospital stay. Her school library subscription covers four databases.~~ Which source gives the strongest evidence that a hand-hygiene program reduces infections?

Target word: strongest evidence **Decides:** Study design ranking.

A	Adjacent category	Strong, but one trial ranks below a review that pools several.
B	Absolute language	Recency matters for currency but does not outrank design.
C	Correct	It pools multiple RCTs and sits at the top of the hierarchy.
D	Confusion set	Observational designs rank below randomized evidence.

4. Answer A Concept 2 • transfer

For a question about whether a treatment works, which source ranks lowest on the evidence hierarchy?

Target word: lowest rank **Decides:** Opinion without systematic data.

A	Correct	Opinion not based on systematic study sits at the bottom.
B	Inverted operation	That is the top of the hierarchy, not the bottom.
C	Adjacent category	RCTs are near the top.
D	Confusion set	Cohort studies are observational but still rank above opinion.

5. Answer A Concept 3

A student at a university ~~with a large online program~~ is researching teen vaping. ~~Her roommate recommends posts by a popular wellness influencer.~~ Which source is most appropriate for an evidence-based capstone?

Target word: most credible source **Decides:** Peer-reviewed, current, unbiased.

A	Correct	It is peer-reviewed, current, and research-based.
B	Dropped element	Drops currency; vaping has changed a great deal since then.
C	Confusion set	A sponsor with a financial interest is a biased source.
D	Camouflage	Popularity is not credibility.

6. Answer D Concept 4

A one-time survey of 2,000 students at a university ~~known for its football team~~ finds that students who sleep less report higher stress. ~~The survey was funded by a campus wellness grant.~~ What conclusion does the design support?

Target word: supported conclusion **Decides:** Cross-sectional design shows association only.

A	Wrong stem number	The sample is large; the limit is the design, not the size.
B	Inverted operation	Reverses the direction but still claims cause without evidence.
C	Absolute language	Claims cause the design cannot support.
D	Correct	A single time point cannot establish temporal order or cause.

7. Answer D Concept 5

A student who volunteers at a food bank ~~with a new walk-in freezer~~ wants to understand how rural caregivers experience arranging transport to dialysis. ~~Her professor prefers statistical studies.~~ Which approach fits the question?

Target word: method fit **Decides:** The question is about lived experience.

A	Adjacent category	A trial tests an intervention's effect, not experiences.
B	Confusion set	Rating scales count, but they miss the depth of experience asked about.
C	Camouflage	The question, not the professor's preference, decides the method.
D	Correct	Experiences and meaning are explored qualitatively.

8. Answer B Concept 5 • transfer

A student wants to know whether average patient wait times differ between two clinics. Which approach fits?

Target word: method fit **Decides:** Comparing amounts calls for numbers.

A	Absolute language	Method must match the question.
B	Correct	A difference in amounts is a quantitative question.
C	Confusion set	Feelings are qualitative; the question is about measured time.
D	Inverted operation	That explores meaning, the opposite type of question.

9. Answer B Concept 6

A capstone student at a hospital ~~that recently opened a new cancer wing~~ wants to survey patients about their care. She has analysis software and ~~a faculty mentor with 30 years' experience~~. Before collecting any patient data, what must she obtain?

Target word: required before data collection **Decides:** Human-subjects research requires review and consent.

A	Adjacent category	Marketing approval is not ethics review.
B	Correct	Research with people requires ethics review and voluntary consent.
C	Dropped element	Drops the IRB and participant consent.
D	Absolute language	Surveys of patients are human-subjects research.

10. Answer D Concept 7

A student writing her literature review ~~during a semester when she also started a new job~~ lists 15 articles one by one, with a paragraph on each. Her instructor says the review needs synthesis. What should she do?

Target word: how to synthesize **Decides:** Synthesis groups and compares across studies.

A	Inverted operation	Synthesis weighs conflicting findings rather than hiding them.
B	Adjacent category	More summaries is more volume, not synthesis.
C	Confusion set	Still a list of summaries.
D	Correct	That is what synthesis means.

11. Answer A Concept 8

~~Using the library's new discovery interface,~~ a student searches "obesity AND children AND school lunch" and gets 40 results. She wants more results, including articles that use the word "pediatric." What should she change?

Target word: how to broaden **Decides:** OR adds synonyms.

A	Correct	OR broadens by accepting either synonym.
B	Absolute language	OR increases results.
C	Inverted operation	AND narrows, returning fewer results.
D	Confusion set	NOT excludes the term she wants.

12. Answer C Concept 8 • transfer

A search for "telehealth OR telemedicine" returns 900 articles. "Telehealth" alone returns 500 and "telemedicine" alone returns 600. How many articles contain both terms?

Target word: overlap count **Decides:** OR total = A + B – both.

A	Dropped element	Subtracts only telehealth from the OR total, dropping the other term.
B	Wrong stem number	Adds the two counts without removing the overlap.
C	Correct	$500 + 600 - 900 = 200$.
D	Confusion set	Takes the difference between the two single-term counts.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	5	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	PICO research questions	A focused clinical question names the Population, Intervention, Comparison, and Outcome (plus Time when used).	Treating the time frame as the comparison, or thinking any question about a topic is researchable.	A draft question checked for missing parts.
2	Hierarchy of evidence	For effectiveness questions, systematic reviews and meta-analyses of randomized trials rank highest, then single RCTs, cohort, case-control, case series, and expert opinion.	Ranking by prestige, recency, or sample size instead of study design.	"Which source is the strongest evidence?"
3	Source credibility	Use peer-reviewed, current, authoritative, unbiased sources for an evidence-based project.	Accepting popular, outdated, or sponsor-produced sources because they are easy to find.	A mix of journal articles, websites, and social media posts.
4	Design limits what you can claim	Cross-sectional and other observational designs show association, not cause, and cannot establish which came first.	Reading a one-time survey correlation as proof of cause.	"At a single point in time" plus a causal conclusion.
5	Qualitative vs. quantitative fit	Questions about experiences and meaning call for qualitative methods; questions about amounts, differences, or effects call for quantitative methods.	Choosing a method by preference instead of by the question.	Words like "experience" and "understand" versus "measure" and "compare".
6	Research ethics and consent	Research with people requires IRB review and voluntary informed consent before data collection, with extra protection for vulnerable groups.	Thinking a mentor's approval or an anonymous survey removes the need for review.	A plan to collect data from patients or students.
7	Synthesis vs. summary	A literature review synthesizes by organizing studies into themes, comparing findings, and naming gaps.	Listing article summaries one by one and calling it a review.	Feedback that a review "needs synthesis".
8	Boolean search logic	AND narrows a search, OR broadens it with synonyms, and NOT excludes a term; OR total = A + B – both.	Adding a synonym with AND and expecting more results.	A need for more (or fewer) search results.

Course scope (public description): The Health Sciences Capstone is a culminating experience for the BS in Health Sciences program. In this project-based course, students will apply their skills obtained through the program to an issue of personal interest in the health sciences. This course requires students to think deeply and use their creative problem-solving skills and understanding of diverse perspectives to identify a research topic and conduct research on it. Upon completion of the capstone, students will have proposed an evidence-based strategy to address a real-world issue related to the health sciences. Students must complete all other courses in this program before attempting the capstone. This course is eligible for an In Progress grade. Please see the Grading

Scale Policy for more information.